

Sailor Shift Q2

Sub-question 1: Was this influence intermittent or did it have a gradual rise?

Visualisation design for both charts

I built three linked views because no single chart answered the question well on its own (Roberts, 2007). I started with a stacked area chart for an overview-first reading of volume and genre mix (Shneiderman, 1996), but stacking makes it hard to compare any genre that isn't sitting on the baseline (Heer, Bostock & Ogievetsky, 2010), so I added small multiples (Tufte, 1983) so each genre could be read on its own scale. The headline view is a streamgraph using Byron and Wattenberg's (2008) wiggle offset, paired with a 3-year rolling average to surface the underlying trend (the build-up of a 'cultural wave') rather than yearly spikes.

Results

The first stacked area chart plots raw counts. This view highlights highly intermittent behaviour, revealing sharp, isolated spikes in 2017, 2023, and 2029. Conversely, the streamgraph applies a three-year rolling average. Visualisation theorist Colin Ware (2012) notes that data smoothing reduces visual noise to expose underlying macro-trends. By smoothing the data, this second view clearly illustrates a gradual, cumulative rise in overall influence building from 2019. Ultimately, these short-term intermittent bursts collectively form a gradual, long-term cultural wave.

```
STYLE = ▶ Object {font: Object, color: Object}
```

```
data = ▶ Array(149) [Object, Object, Object, Object, Object, Object, Object, Objec
```

```
data = FileAttachment("q1_data.csv").csv({typed: true})
processedData = ▶ Object {data: Array(224), genres: Array(), minYear: 2007, maxYe
```

```
ftColors = ▶ Object {Dream Pop: "#0f5499", Indie Folk: "#990f3d", Desert Rock: "#f
```

Filter genres:

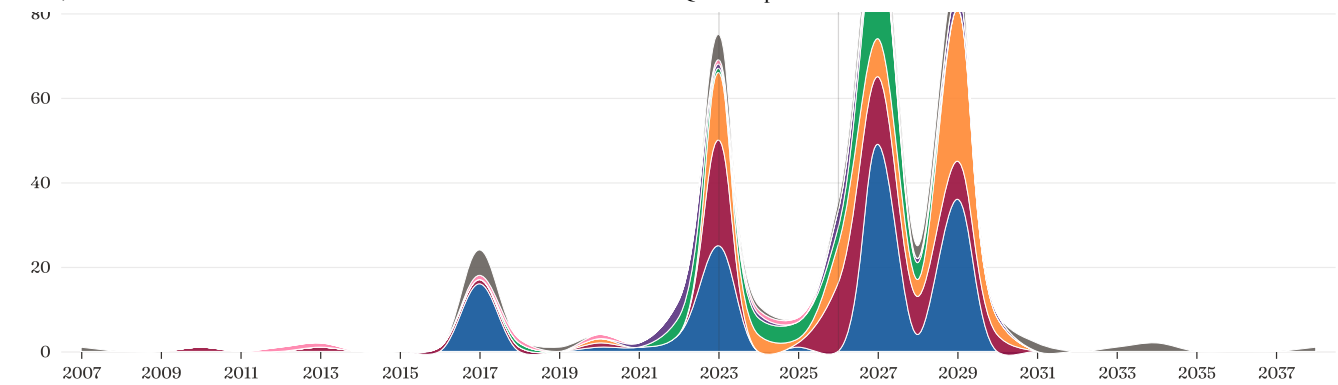
☒ Dream Pop ☒ Indie Folk ☒ Desert Rock ☒ Space Rock ☒ Synthwave ☒ Americana
☒ Other

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Overview of Oceanus Folk influence

Total influenced works per year, organised by genre. Toggle genres above to focus the view.





Select view:

☒ Streamgraph ☐ Small multiples

selectedGenre = null

Show all

Dream Pop

Indie Folk

Desert Rock

Space Rock

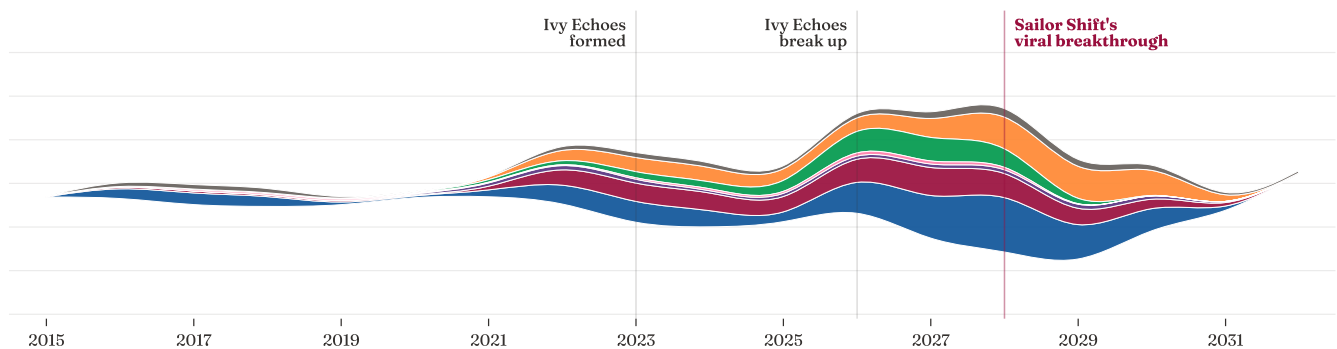
Synthwave

Americana

Other

The cultural wave preceded the commercial breakthrough

Streamgraph values are 5-year rolling averages. Small multiples view shows raw counts.



Sub-question 2: What genres and top artists have been most influenced by Oceanus Folk?

Visualisation design (Sankey diagram)

I selected a Sankey diagram to map the trajectory of influence, following it from Oceanus Folk through individual genres and finally to work types (Songs or Albums). I mapped ribbon width to the volume of influenced works, since it is an accurate channel for quantitative perception, it allows for clear comparisons of influence flow (Munzner, 2014). To handle the high density, I implemented a click-to-expand interaction that adheres to Shneiderman's (1996) "details-on-demand" principle. This allows the user to investigate the top items within a genre without overwhelming the initial view. By aggregating the long tail of smaller genres, the chart maintains clarity and scales effectively as the dataset grows.

Results (Sankey diagram)

The Sankey diagram illustrates the genre influence trajectory of 149 total works. The

thickest flow lines clearly reveal the most influenced genres. These patterns show Dream Pop (29 works), Indie Folk (28), and Desert Rock (24) as the top categories. Furthermore, the right side of the flow splits to show this influence occurs predominantly through individual songs (115) rather than full albums (34).

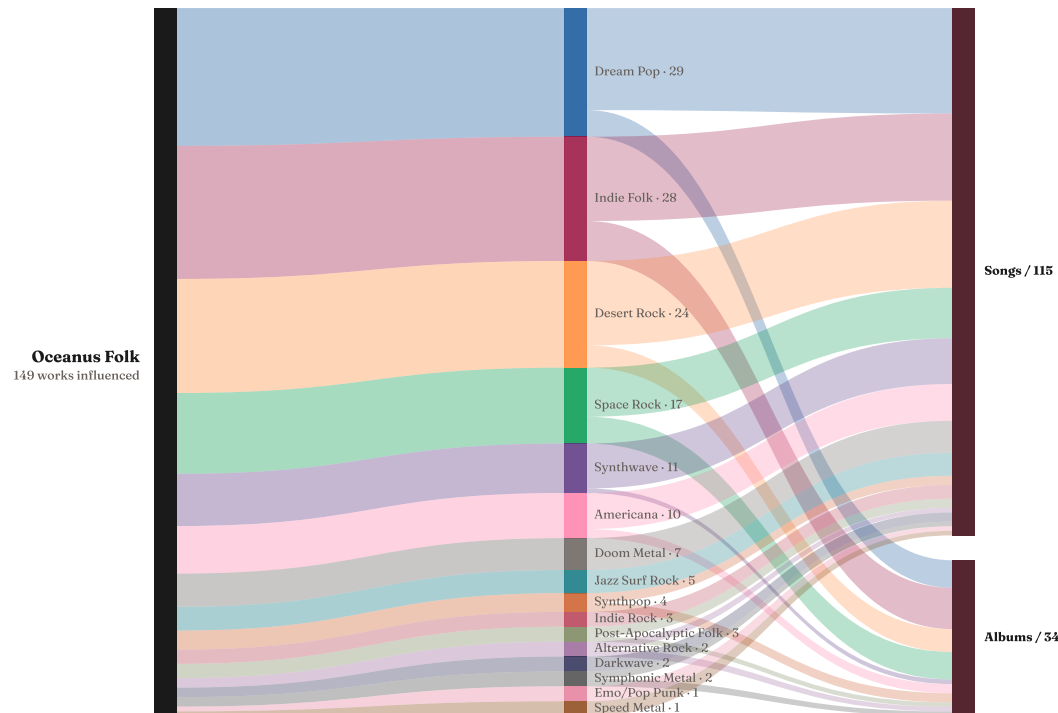
```
q2data = ► Array(247) [Object, Object, Object, Object, Object, Object, Object, Object, Obj

sankeyData = ► Object {genres: Array(16), totalSongs: 115, totalAlbums: 34, totalW

expandedGenre = null
```

Genre influence trajectory

Click a genre to expand into its top songs and albums. Click again to collapse.



```
q2p2data = ► Array(1144) [Object, Object, Object, Object, Object, Object, Object,

artistNetwork = ► Object {artists: Array(602), totalArtists: 602}
```

Choose the number of top artists:

☒ 15 ☐ 25 ☐ 40

```
expandedArtist = null
```

Visualisation design (Radial ego-network diagram)

I used a radial ego-network to explore the Oceanus Folk's impact on individual

collaborators. By placing Oceanus at the centre of a ring, I established a clear visual

```
mutable expandedGenre = null
```

This radial layout is more stable than a force-directed graph, preventing the visual 'hairball' effect often seen with numerous nodes and edges. Combined with zoom, the choice of the 'number of top artists' shown and pan interactions, this ensures the tool remains responsive and navigable.

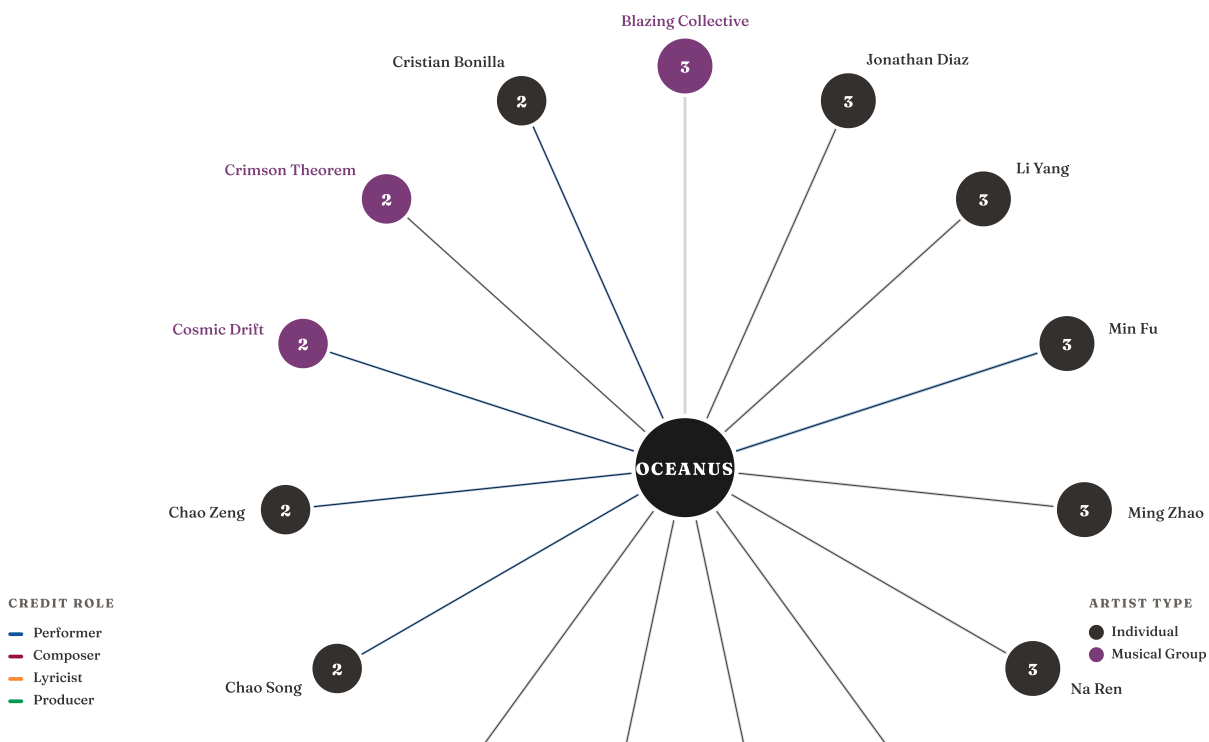
Results (Radial ego-network diagram)

Ten artists share the top position with three works each, including Blazing Collective and Jonathan Diaz. The categorical colour coding further reveals that these top artists are primarily individuals (black nodes) acting simultaneously multiple roles of composers, performers, lyricists, producers (grey links - details about each roles and associated works are available upon clicking on the nodes).

The top artists influenced by Oceanus Folk

[RESET VIEW](#)

Node size scales with number of influenced works. Click an artist to see their songs, albums and roles.



Sub-question 1: Was this influence intermittent or did it have a gradual rise?

Visualisation design

A chord diagram makes it easy to see how genres 'blend' into Oceanus Folk, hence highlighting their influence on it. I used ribbon width to encode the volume of influenced works as these are precise visual channels for quantitative data (Munzner 2014). The

```
mutable expandedArtist = null
```

added filtering of 'eras' - before Sailor, during, after - to allow users to compare how the influence has changed over time. The circular design of the chord diagram allows for an easy visual comparison of these changes.

Results

The major pattern shows a reduction and fragmentation of inspiration following the Sailor Shift breakthrough. In the 'Legacy' era, Oceanus Folk drew heavily from Indie Folk (17 works) and Synthwave (15). In 'Emergence' one, from Dream Pop (7 works), Indie Folk (5 works), Doom Metal (4 works). Conversely, in the 'Contemporary' era, this dominant reliance on certain genres completely disappears. Contemporary inspiration is highly fragmented and minimal. It now draws only one or two works evenly across a wider spread of genres, primarily Desert Rock, Dream Pop, and Space Rock.

```
q3data = ►Array(118) [Object, Object, Object, Object, Object, Object, Object, Object, Obj
```

```
q3_matrix = ►Array(118) [Object, Object, Object, Object, Object, Object, Object, Object,
```

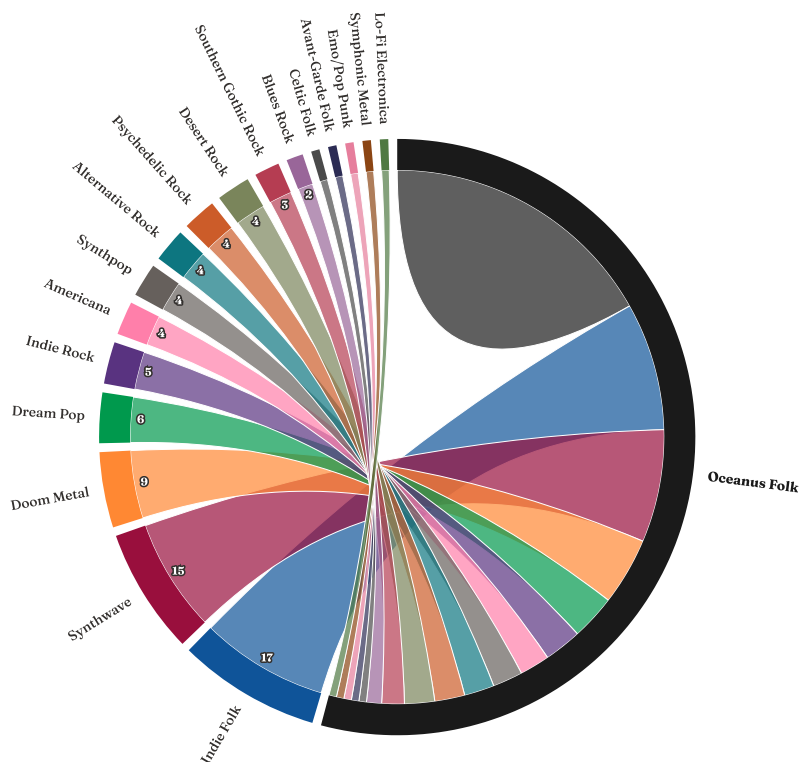
```
eras = ►Array(3) [Object, Object, Object]
```

Choose era:

Legacy (before 2023) ▼

Where does Oceanus Folk draw its inspiration?

Pre-Sailor Shift, Legacy era. 1981 to 2022, Oceanus Folk's roots.



TOP INSPIRATIONS

Reflection

First, I struggled to visualise temporal data effectively. A multi-line chart initially seemed appropriate to show exact volumes without distortion. However, overlapping lines caused severe visual clutter, and raw yearly data obscured long-term trends. I solved this using linked views. I combined a streamgraph using a three-year rolling average to reveal the macro-trend with small multiples to show exact, independent genre counts accurately.

Second, mapping artist influence initially created a cluttered network 'hairball'. I resolved this by replacing a standard force-directed graph with a radial ego-network. Anchoring Oceanus Folk at the centre and filtering for top artists maintained structural hierarchy without visually overwhelming the user.

Also reflecting on my data analysis, relying solely on absolute counts limits the findings. I did not compare influenced works against an artist's or genre's total overall output. Calculating the percentage of an artist's total discography influenced by Oceanus Folk would provide a much more accurate, relative ranking of true impact.

To make these complex visualization challenges fundamentally easier to resolve without requiring to manually engineer custom workarounds, future research should focus on AI-Driven Visual Analytics (Wu et al., 2021). To navigate the complexities of working with big network graphs, future visualization libraries could arguably integrate machine learning models that automatically monitor visual clutter and data density in real-time. If a network degrades into a hairball, the system would autonomously detect the cognitive overload and instantly apply the optimal decluttering technique—intelligently aggregating minor genres or simplifying network structures based on what the user is trying to explore.

References

Ware, C. (2012) Information visualization: perception for design. 3rd edn. Waltham, MA: Morgan Kaufmann.

Byron, L. and Wattenberg, M. (2008). Stacked Graphs – Geometry & Aesthetics. IEEE Transactions on Visualization and Computer Graphics, 14(6), pp. 1245-1252.

Heer, J., Bostock, M. and Ogievetsky, V. (2010). A tour through the visualization zoo. Communications of the ACM, 53(4), pp. 59-67.

Munzner, T. (2014). Visualization Analysis and Design. A K Peters/CRC Press.

Roberts, J.C. (2007). State of the Art: Coordinated & Multiple Views in Exploratory Visualization. Fifth International Conference on Coordinated and Multiple Views in Exploratory Visualization (CMV 2007), pp. 61-71.

Shneiderman, B. (1996). The Eyes Have It: A Task by Data Type Taxonomy for Information Visualizations. IEEE Symposium on Visual Languages, pp. 336-343.

Tufte, E.R. (1983). The Visual Display of Quantitative Information. Graphics Press.

Wu, A., et al. (2021). AI4VIS: Survey on Artificial Intelligence Approaches for Data Visualization. IEEE Transactions on Visualization and Computer Graphics, 28(1), 512-522.

```
q3data = FileAttachment("genre_influence_timeline.csv").csv({typed: true})
```